

STEP Interface Functional Specification

Ngee Ann Polytechnic – Library Systems



Version 1.0

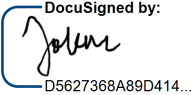
Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	26/09/2023	Initial version of NP library systems IFS	AvePoint
0.2	11/10/2023	2.1 System Flow table SN 3 description update	AvePoint
1.0	11/10/2023	Sign-off version	AvePoint

Document Acceptance Form

Prepared By:

Jolene Oon
Business Analyst, AvePoint

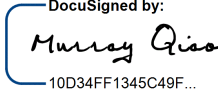
Signature: 
Date:

Review By:

Mint Lang
Solution Architect, AvePoint

Signature: 
Date:

Murray Qiao
Technical Manager, AvePoint

Signature: 
Date:

Approved By:

Andrew Tan
Assistant Director, Academic Quality
& Shared Services, CET Academy,
Ngee Ann Polytechnic (NP)

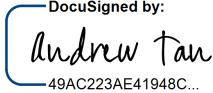
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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS Skills Training Enhancement Portal (STEP) and Ngee Ann Library Systems to interface students' data for the creation of Continuing Education & Training (CET) students' library accounts and student can access the resource of the library.

STEP shall identify students' details based on the CIS Student ID, enrolment information, course end date and personal information, such as name and email address, to generate corresponding XML files. STEP shall generate and deposit the XML files encrypted by PGP protocol in the STEP SFTP server accordingly. Ngee Ann Polytechnic can create student library account and grant gantry access based on the transferred files.

1.2 Intended Audience

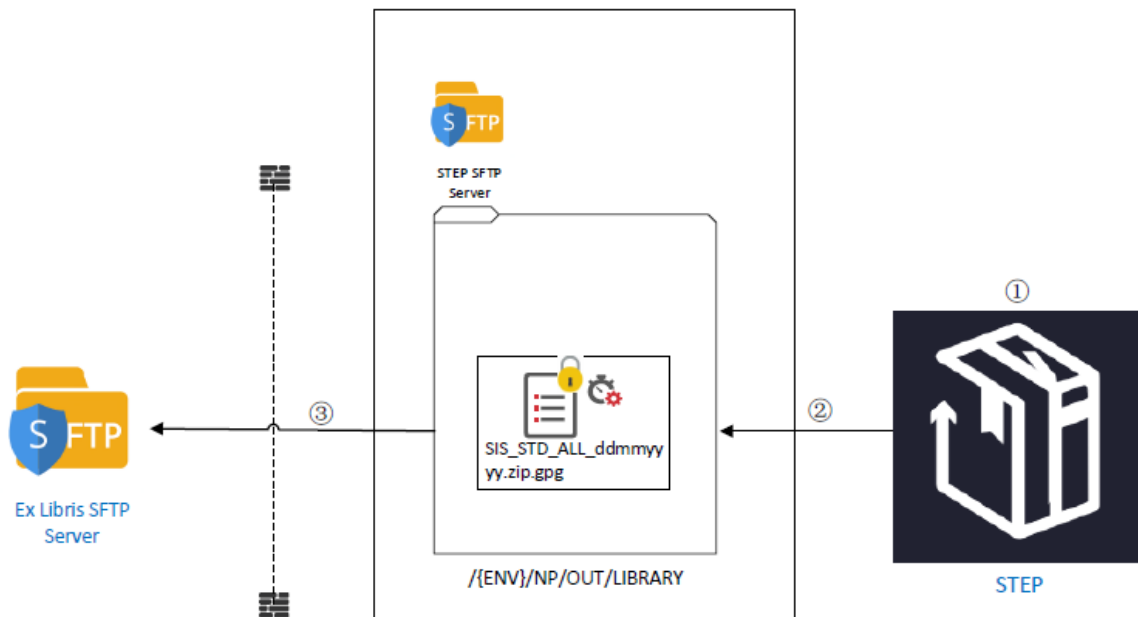
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of NP Library System and STEP.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **Ex Libris SFTP Server:** Ngee Ann Polytechnic library system vendor, Ex Libris SFTP server as an alternative to using institution's own FTP solutions.
- **STEP SFTP Server:** A SFTP server hosted in Azure that stores encrypted file used for data exchange.
- **STEP:** Skills Training Enhancement Portal



SN	Description
1	STEP generates the XML files based on the student's enrollment status, the student only enrolled in FQ and some short courses will be exported. Please refer to TP FQ & Non-FQ list here .

2	STEP configures daily timer job push to STEP SFTP server after zip the XML files and encryption with PGP protocol.
3	STEP push file into Ex Libris SFTP server for downstream system to consume data.

2.2 Interface Design Summary

Output Interface File

Interface File Name	STU_(running no.).xml; e.g. STU_1.xml, STU_2.xml XML file passed must always be in zip folder format. E.g. SIS_STD_All_ddmmyyyy.zip
Data Source System	Skills Training Enhancement Portal
Data Destination System	Ngee Ann Polytechnic Library Systems
Mode of Interface	XML file
Method	SFTP
File Path	/{ENV}/NP/OUT/LIBRARY ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Mode	full load data sync
Frequency	Daily
Timing	Configurable
File Retention Policy	Files are retained for a maximum of 30 days in the STEP SFTP server, after this period the files will be automatically deleted
Possible Exception and Exception Handling	1. File loss or transfer failure due to STEP internal exception. E.g., Timer service down, etc. Exception handling steps: - STEP support team shall be informed by email which sent by STEP of the transfer failure. - AP team checks the relevant logs to troubleshoot the cause of the failure. - If the issue is due to the transfer file could not be found in STEP SFTP server: - STEP support team will manually rerun the timer job to trigger file generation and deposit the file in STEP SFTP server.

	2. Data error. Exception handling steps: - STEP support team shall be informed by email which sent by STEP of the file generation failure. - AP team checks the relevant logs and file to troubleshoot the cause of the failure (data error types are listed below). - Field is mandatory but missing - Field failed validation - AP redeposits the correct file in SFTP.
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2.3 Business Rule

Ngee Ann Polytechnic (NP) students who are enrolled into full qualification courses would be allowed to access NP's library resources.

STEP system to send .xml file in zip folder as mapped in *Table 3.0: Fields Mapping* to NP library system for student access management. The scope and definitions are illustrated as in *Table 1.0: Student Status's Mapping* and *Table 2.0: Full Qualification Courses Scope*.

Table 1.0: Student's Status Mapping

No.	Student's status in		Single intake	Multiple intakes
	STEP	NP library		
1	Pending enrolment	<i>Do not include</i> in data file		
2	Transferred			
3	Enrolled	Active	Include in data file. {expiry date} is course intake's end date	Include in data file. {expiry date} is course intakes' latest end date
4	Ready to complete			
5	Ready to graduate			
6	Withdrawn	Inactive	Include in data file until {expiry date} + 1 day only.	

7	Deferred		{expiry date} is yesterday.
8	Suspended		
9	Completed		
10	Graduated		

Note: If a student has multiple statuses such as “Withdrawn”, “Enrolled” and “Transferred”, student’s {expiry date} will be the “Enrolled” status course intakes latest end date.

Table 2.0: Full Qualification Courses Scope

Course Type	Code	Remark(s)
Academic Full Qualification	FQ	FQ
Academic Certificate	MC	FQ
Academic Modular	SBMC	FQ
Work-Study Certificates Leading To Part-Time Diploma	WSC2PTD	FQ
Work-Study Post-Diploma Specialist Diploma	WSPSD	FQ
Work-Study Post-Diploma Leading To Specialist Diploma	WSP2SD	FQ
Work-Study Post-Diploma Advanced Diploma	WSPAD	FQ
Work-Study Post-Diploma Diploma Conversion	WSDC	FQ
Critical Core Skills	CCS	Non-FQ

2.4 Interface File Generation Rule

Flow to Generate Library Data

SN	Description
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1	STEP identifies the student's enrollment status daily. Once the student is enrolled, STEP shall locate the corresponding student information and course information. Refer to 3. Interface Fields for the fields to be sent.
2	STEP generates XML files, containing up to 500 student records per .xml file to consolidate the full-load data of students.
3	STEP deposits the XML files in zip format in SFTP server accordingly. SFTP Server Hostname: custdata-dc05.hosted.exlibrisgroup.com SFTP Server Port No.: 10022

3. Interface Fields

Student information for library account creation:

.xml File Fields	Value in STEP	Description	Data Type	Remarks
record_type	PUBLIC		String(\$nvarchar(6))	Default to "PUBLIC"
primary_id	Student ID	The student ID assigned to each student to login STEP system in alphanumerical format.	String(\$nvarchar(50))	Sample value: 1234567A
last_name	Student name	Student's name as in STEP in upper case letter style.	String(\$nvarchar(50))	Sample value: STEP TEST PERSON
user_title	MR/MS	User's title by gender identified in student's profile.	String(\$nvarchar(2))	Sample value: MS
user_group	STUDENT		String(\$nvarchar(2))	Default to "STUDENT"
cataloger_level	00		String(\$nvarchar(2))	Default to "00"
preferred_language	en		String(\$nvarchar(2))	Default to "en"
expiry_date	Course intake end date	The end date of a course intake in STEP in UTC timezone. Refer to business rule for detailed information.	String(\$nvarchar(10))	Format: YYYY-MM-DDZ

account_type	EXTERNAL		String(\$nvarchar(8))	Default to "EXTERNAL"
external_id	SIS		String(\$nvarchar(8))	Default to "SIS"
force_password_change	false		String(\$nvarchar(8))	Default to "false"
status	ACTIVE		String(\$nvarchar(6))	Default to "ACTIVE"
<Contact_info>email_address</Contact_info>	Student's personal email address		String(\$nvarchar(50))	Sample value: steptestperson@gmail.com
<Contact_info>email_type</Contact_info>	personal		String(\$nvarchar(10))	Default to "personal"
<user_identifiers>id_type</user_identifiers>	OTHER_ID_1		String(\$nvarchar(10))	Default to "OTHER_ID_1"
<user_identifiers>Value</user_identifiers>	Student ID	Student's identification used to login to STEP in full.	String(\$nvarchar(50))	Sample value: 1234567A@polite.edu.sg
<user_identifiers>Status</user_identifiers>	ACTIVE		String(\$nvarchar(6))	Default to "ACTIVE"
<user_roles>Status</user_roles>	ACTIVE		String(\$nvarchar(6))	Default to "ACTIVE"
<user_roles>Scope</user_roles>	65SPO_NP		String(\$nvarchar(10))	Default to "65SPO_NP"
<user_roles>role_type</user_roles>	200		String(\$nvarchar(3))	Default to "200"
<user_note>note_type</user_note>	BARCODE		String(\$nvarchar(10))	Default to "BARCODE"

<user_note> note_text </user_note>	ACADPROG: {diploma info if any}	ACADPROG: {course code}	String(\$nvarchar(50))	Sample Value: ACADPROG:NP-DEMH
<user_note> user_viewable </user_note>	FALSE		String(\$nvarchar(10))	Default to "FALSE"
<user_note> popup_note </user_note>	FALSE		String(\$nvarchar(10))	Default to "FALSE"
<user_note> created_by </user_note>	System		String(\$nvarchar(10))	Default to "System"
<user_note> created_date </user_note>	File created date and time	The date and time file was created in UTC timezone.	String(\$nvarchar(50))	Sample value: 2023-06-19T10:59:00Z
statistic_category	NP Full Qualification Student		String(\$nvarchar(50))	Default to "NP Full Qualification Student"
category_type	USERPROFILE		String(\$nvarchar(50))	Default to "USERPROFILE"
statistic_category	ACADCAREER: FT		String(\$nvarchar(50))	Default to "ACADCAREER: FT" There is a 'space' before FT.
category_type	STUDENTTYPE		String(\$nvarchar(50))	Default to "STUDENTTYPE"
statistic_category	DEPTSCHOOL: CET	DEPTSCHOOL: {school}		Sample value: DEPTSCHOOL: FMS If there is multiple school in course intake, only pick up the first school code. There is a 'space' before school code.



category_type	SCH_DEPT		String(\$nvarchar(50))	Default to "SCH_DEPT"
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4. Sample Files



STD_1.xml